MOI GIRLS' ELDORET 2025 PREDICTION

POSSIBLE KCSE EXAM

BIOLOGY

231/3

PAPER 3

Practical

TIME: 1¾ HOURS

NAME…………………………………………………………………………………

SCHOOL……………………………………………… **SIGN**………………………

INDEX NO…………………………………………… **ADM NO**……………………

(Kenya Certificate of Secondary Education)

Instructions to candidate

- *Answer ALL questions*
- *You are required to spend the first 15 min of 1¾ hours allowed for this paper reading the whole paper before carefully before commencing your work.*
- *Answer must be written in the spaces provided in the question paper*
- *Don't insert additional page /paper*

FOR EXAMINER USE ONLY

QUESTIONS	MAXIMUM SCORE	CANDIDATE SCORE
1	14	
2	13	
3	13	
TOTAL	40	

<u>Answer ALL questions</u>

1. You are provided with specimen A and B

a) Name the sub-division to which the specimen belong **(1mk)**

……

b) Name the class to which the specimens belong **(2mks)**

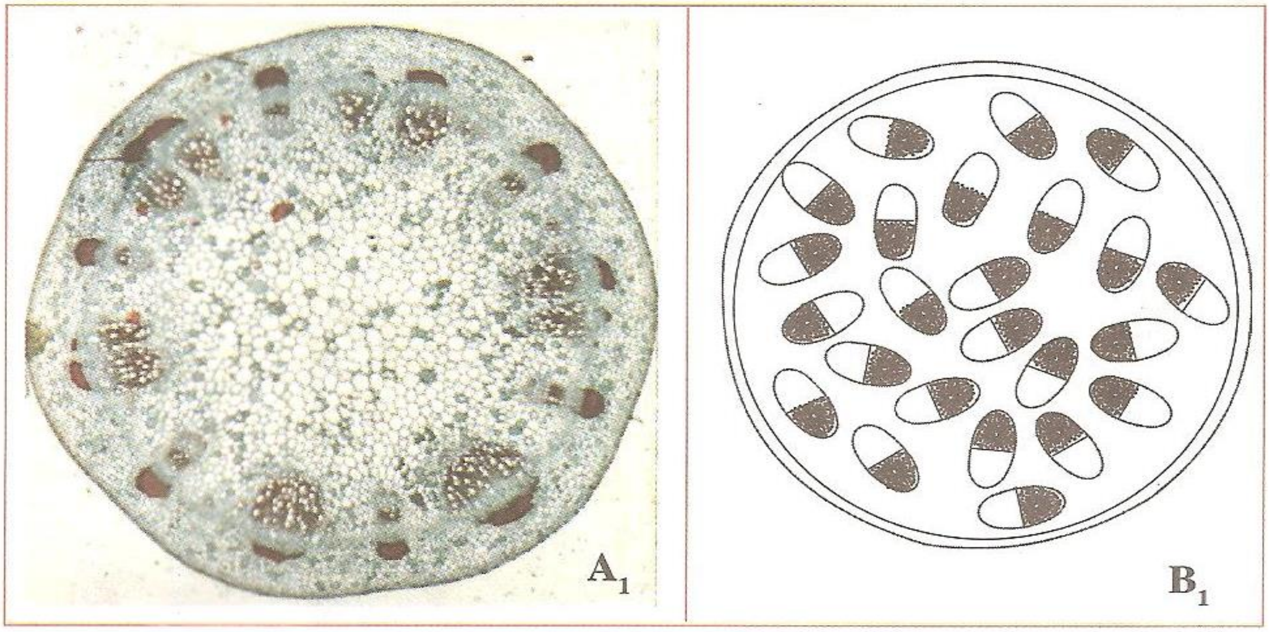
A…………………………………………..

B…………………………………………..

c) State three observable differences between the leaves of specimen A and B **(3mks)**

Leaves A	**Leaves B**

d) The diagram below show the cross-sections of stems obtained from specimens A and B



(i) Match the stem cross-sections with the specimens **(2mks)**

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(ii) Outline three differences between the two stems **(3mks)**

Specimen A_1	Specimen B_1

e) Suggest the agent of pollination of the flowers of specimen A **(1mk)**

……………………………………………………………………………………………………

Give reason for your answer **(1mk)**

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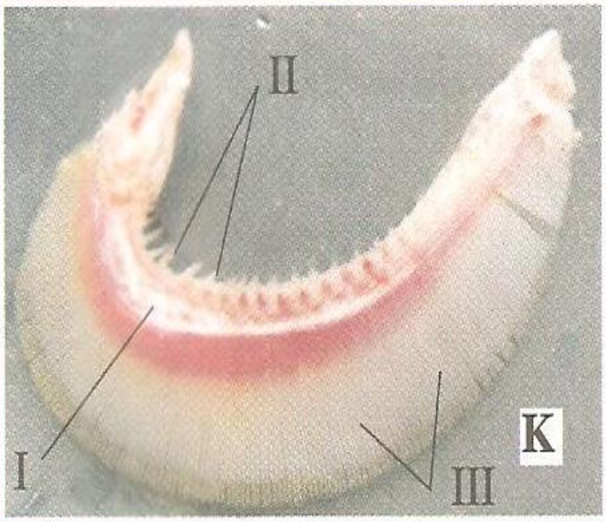
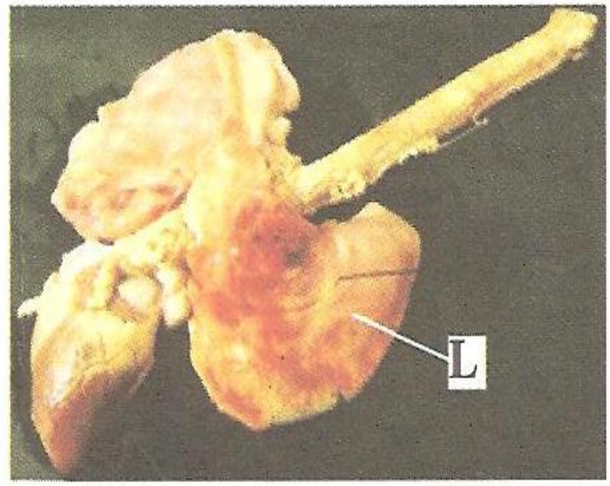
2. You are provided with photographs of specimens labeled **K (gills)** and **L (lungs).** Examine them and answer the questions that follow.





(a) Name the class of organisms from where the specimens were obtained **(2mks)**

Specimen	**Class**
K	
L	

(b) Label all the parts of specimen K on the photograph **(3mks)**

I…………………………

II………………………..

III………………………

(c) State the functions of each of the parts you have labeled in (b) above **(3mks)**

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(d) State three ways in which the part labeled L is adapted to its functions **(6mks)**

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(e) State the functional relationship between specimens K and L **(1mk)**

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3. You are provided with:

- 1ml Olive oil
- K_1 (Concentrated sodium hydrogen carbonate solution)
- K_2
- Irish potato
- Test tube
- Iodine solution

Label two test tubes X and Y. Into each test tube; put 2cm^3 of water and 8 drops of Olive oil. To the test tube labeled X, add 8 drops of Liquid K_1. Shake both test tubes and allow the contents to stand for 2 minutes

a) (i) Record your observation in: **(2mks)**

Test tube X

……………………………………………………………………………………………………

……………………………………………………………………………………………………

Test tube Y

……………………………………………………………………………………………………

……………………………………………………………………………………………………

(ii) Name the process that has taken place in test tube X **(1mk)**

…………………………………………………………………………………………………

(iii) Stage the significance of the process named in (a) (i) above in digestion **(1mk)**

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…………………………………………………………………………………………………

(iv) Name the:

I. Digestive juice in human beings that has the same effect on oil as liquid K_1 **(1mk)**

……………………………………………………………………………………

II. Region of alimentary canal where the juice is secreted **(1mk)**

………………………………………………………………………………………

b) Label two test tubes E and F. place $2cm^3$ of liquid K_2 into each. Add a drop of iodine solution into each test tube

(i) Record your observations **(1mk)**

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…………………………………………………………………………………………………

(ii) Suggest the identity of liquid K_2 **(1mks)**

…………………………………………………………………………………………………

(iii) Cut out a cube whose sides are 1cm from the irish potato provided. Crush the cube to obtain a paste and place the paste in the test tube labeled E. Leave the set up for at least 30minutes

Record your observations **(2mks)**

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(iv) Account for the results in (b) (iii) above **(2mks)**

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